

1. Reconstruir 23D

No necesitamos tocar nada anterior.

DROP TABLE IF EXISTS

bu_frd.model_behavior_features_current_2022_2025;

CREATE TABLE

bu_frd.model_behavior_features_current_2022_2025

USING PARQUET

AS

SELECT

*,

/* =====
SPARSE EVENTS
===== */

deterioration_recent_3m / 3.0
- deterioration_prior_9m / 9.0
AS deterioration_acceleration,

adverse_recent_3m / 3.0
- adverse_prior_9m / 9.0
AS adverse_acceleration,

/* =====
CONTINUOUS / EXPOSURE CHANGE
===== */

avg_active_accounts_recent_3m
- avg_active_accounts_prior_9m
AS active_accounts_change_vs_baseline,

avg_balance_recent_3m
- avg_balance_prior_9m
AS balance_change_vs_baseline,

avg_past_due_recent_3m
- avg_past_due_prior_9m
AS past_due_change_vs_baseline,

delinquency_share_recent_3m
- delinquency_share_prior_9m

```

    AS delinquency_share_change_vs_baseline,

    avg_utilization_recent_3m
    - avg_utilization_prior_9m
    AS utilization_change_vs_baseline,

    weighted_utilization_recent_3m
    - weighted_utilization_prior_9m
    AS weighted_utilization_change_vs_baseline

FROM
    bu_frd.model_party_behavior_windows_2022_2025;

Después:

SELECT
    COUNT(*) AS rows,
    COUNT(DISTINCT party_id) AS parties,

    SUM(
        CASE
            WHEN deterioration_acceleration IS NULL
            THEN 1 ELSE 0
        END
    ) AS deterioration_missing,

    SUM(
        CASE
            WHEN utilization_change_vs_baseline IS NULL
            THEN 1 ELSE 0
        END
    ) AS utilization_change_missing,

    SUM(
        CASE
            WHEN weighted_utilization_change_vs_baseline IS NULL
            THEN 1 ELSE 0
        END
    ) AS weighted_util_change_missing

FROM
    bu_frd.model_behavior_features_current_2022_2025;

```

Ahora esperamos exactamente:

rows	688,589
parties	688,589
deterioration_missing	0

utilization_change_missing	198,963
weighted_util_change_missing	198,963

2. Hay un último detalle antes de congelar el Z-score

Como corregimos la definición de `weighted_utilization`, yo volvería a calcular únicamente el P99.9 de `weighted utilization` en TRAIN.

No porque espere un cambio grande, sino porque el threshold:

1.75333

fue calculado con la versión anterior.

Para reproducibilidad, el cap del challenger debe provenir de la tabla corregida.

```
SELECT
  percentile_approx(
    weighted_utilization,
    ARRAY(
      0.95,
      0.99,
      0.995,
      0.999
    )
  ) AS weighted_util_train_percentiles,

  MIN(weighted_utilization) AS min_weighted_utilization,
  MAX(weighted_utilization) AS max_weighted_utilization

FROM
  bu_frd.model_party_utilization_monthly_2022_2025 u

INNER JOIN (

  SELECT DISTINCT
    party_id,
    observation_date,
    model_split

  FROM
    bu_frd.model_party_behavior_monthly_2022_2025

) c

ON u.party_id = c.party_id
AND u.observation_date = c.observation_date
```

WHERE

c.model_split = 'TRAIN'

AND u.weighted_utilization IS NOT NULL;

Si vuelve a dar:

$P99.9 \approx 1.75333$

dejamos todo tal cual.

Si cambia, simplemente actualizamos ese threshold antes de construir definitivamente el Z-score.
